



Course Specification

— (Bachelor)

Course Title: **Introduction to robotics**

Course Code: **EE 414**

Program: **Bachelor of Electrical Engineering**

Department: **Electrical Engineering**

College: **College of Engineering**

Institution: **Alfaisal University**

Version: **V 2024**

Last Revision Date:



Table of Contents:

A. General information about the course:	3
B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods	4
C. Course Content	6
D. Students Assessment Activities	6
E. Learning Resources and Facilities	6
F. Assessment of Course Quality	7
G. Specification Approval	7





A. General information about the course:

Course Identification

1. Credit hours: 3 (3-0-0)

2. Course type

a. University ☐ College ☐ Department ☒ Track ☐ Others ☐

b. Required ☐ Elective ☒

3. Level/year at which this course is offered: Fourth Year

4. Course general Description:

This course introduces robotics and its applications, emphasizing foundational concepts like Modeling, control, perception, and planning and navigation.

5. Pre-requirements for this course (if any):

EE 306 Control and Feedback Systems design

6. Co- requirements for this course (if any):

7. Course Main Objective(s):

This course provides a comprehensive understanding of robotics and provides students with practical experience through designing, building, and programming robots. Students will acquire proficiency in modeling and designing controllers needed for actuating the robots. Furthermore, they have the ability to plan and navigate robots to specified goal positions.

2. Teaching mode (mark all that apply)

No	Mode of Instruction	Contact Hours	Percentage
1	Traditional classroom	45	100%
2	E-learning		
3	Hybrid - Traditional classroom - E-learning		
4	Distance learning		

3. Contact Hours (based on the academic semester)

No	Activity	Contact Hours
1.	Lectures	45
2.	Laboratory/Studio	0
3.	Field	0
4.	Tutorial	0
5.	Others (specify)	0
Total		45



B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods

Code	Course Learning Outcomes	Code of CLOs aligned with program	Teaching Strategies	Assessment Methods
1.0	Knowledge and understanding			
1.1	Identify the system model and apply appropriate Control design in order to function the for various Electrical Engineering systems.	PLO 1 (SO 8) Knowledge Base for Engineers	Lectures Discussion Readings	Forum Postings Role-Playing
2.0	Skills			
2.1	Solve, with proper notation, and analyze Robotic systems problems using efficient and effective methods.	PLO 2 (SO 1) Problem-solving	Interactive lecture, Class Discussion Posting solution file Project case study Office hours, Solving quizzes and midterms interactively in the classroom.	Assignments Quizzes Midterm Exams Final Exam Project
2.2		PLO 3 (SO 7) Lifelong Learning	Extra readings Invited speakers Seminar/Webinar	Report Project Final exam
2.3	Design feedback controllers and estimators in robotic system to meet specified requirements while assessing performance and resource constraints. This includes designed of full navigation system for robot localization and motion planning.	PLO 4 (SO 2) Creativity & Design	Modelling Discussion	Project Exercise Final exam
2.4		PLO 5 (SO 6) Use of Engineering Tools	Experimentation Hands-on practice	Project Exercise
2.5		PLO 6 (SO 3) Communication & ICT	Sample reports and presentations	Report Presentation
3.0	Values, autonomy, and responsibility			
3.1		PLO 7 (SO 4) Values & Ethics	Discussions Readings Assessment Feedback	Reports Final exam
3.2		PLO 8 (SO 5) Autonomy & Responsibility	Discussions Role Play	Presentation Teamwork

Not all SO attainment results need to be reported in the PLO attainment sheet, only selected mappings as per the Program Specifications mappings.





C. Course Content

No	List of Topics	Contact Hours
1.	Introduction to Robotics	6
2.	Modeling (Inverse kinematics, system identification)	9
3.	Feedback Control and estimation	6
4.	Machine learning approaches to Robot Control	6
5.	Localization and Mapping	6
6.	Motion Planning	6
7.	Robot Navigation	6
Total		45

D. Students Assessment Activities

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.	Attendance and participation	Across the semester	5%
2.	Assignments	Across the semester	10%
3.	Quizzes	Across the semester	10%
4.	Midterm Exam I	Week 6 or 7	15%
5.	Midterm Exam II	Week 12 or 13	15%
6.	Project	Week 14	15 %
7.	Final Examination	Final Exam Week	30%
8.	Total		100 %

*Assessment Activities (i.e., Written test, oral test, oral presentation, group project, essay, etc.)

E. Learning Resources and Facilities

1. References and Learning Resources

Essential References	Kelly, A, Mobile Robotics: Mathematics, Models, and Methods, Cambridge
Supportive References	P. Corke, Robotics, Vision and Control, Springer, 2017 K. Lynch and F. Park, Modern Robotics, Cambridge University Press, 2017 S. Thrun, W. Burgard, D. Fox, Probabilistic Robotics, MIT Press, 2006 R. Siegwart, I. Nourbakhsh, D. Scaramuzza, Autonomous Mobile Robots, 2011 N. Correll, B. Hayes, C. Heckman, A. Roncone, Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms, MIT Press, 2021
Electronic Materials	http://elearning.alfaisal.edu/coe/
Other Learning Materials	Course lecture notes





2. Required Facilities and equipment

Items	Resources
facilities (Classrooms, laboratories, exhibition rooms, simulation rooms, etc.)	Classrooms are standardized in quality and efficiency. Labs are available for specific courses.
Technology equipment (projector, smart board, software)	Smartboard, podium, data show, internet-connected computer, and MS Office software. Access to eLearning for attendance, grading, etc.
Other equipment (depending on the nature of the specialty)	Computer labs are accessible to students with appropriate software pre-installed.

F. Assessment of Course Quality

Assessment Areas/Issues	Assessor	Assessment Methods
Effectiveness of teaching	Students	CES (Indirect)
Effectiveness of students' Students assessment	Faculty	Direct
Quality of learning resources	Students	CES (Indirect)
The extent to which CLOs have been achieved	Faculty	FCAR (Direct)
Other		

Assessor (Students, Faculty, Program Leaders, Peer Reviewer, Others (specify))

Assessment Methods (Direct, Indirect)

G. Specification Approval

COUNCIL /COMMITTEE	EE DEPT. COUNCIL
REFERENCE NO.	
DATE	

CREATED BY	DR. ASEM IBRAHIM ALALAWAN
DATE	20-10-2024
LAST EDITED BY	DR. ASEM IBRAHIM ALALAWAN
DATE	

